

Experiment Report — Graduated Advisory-Scope Governance: Specification, Characterisation and Robustness

Prepared for: supervisory discussion **Date:** 2026-09-21 (revised under `final-research-chain-lock.md`) **Status:** Preliminary finding. Principal figures reproduced from canonical scripts on 2026-09-21. **Subject:** Activation frequency of the intermediate advisory-scope state, Δ_{L2} = 5.81% PRIMARY / 4.48% RESOLUTION, and its robustness.

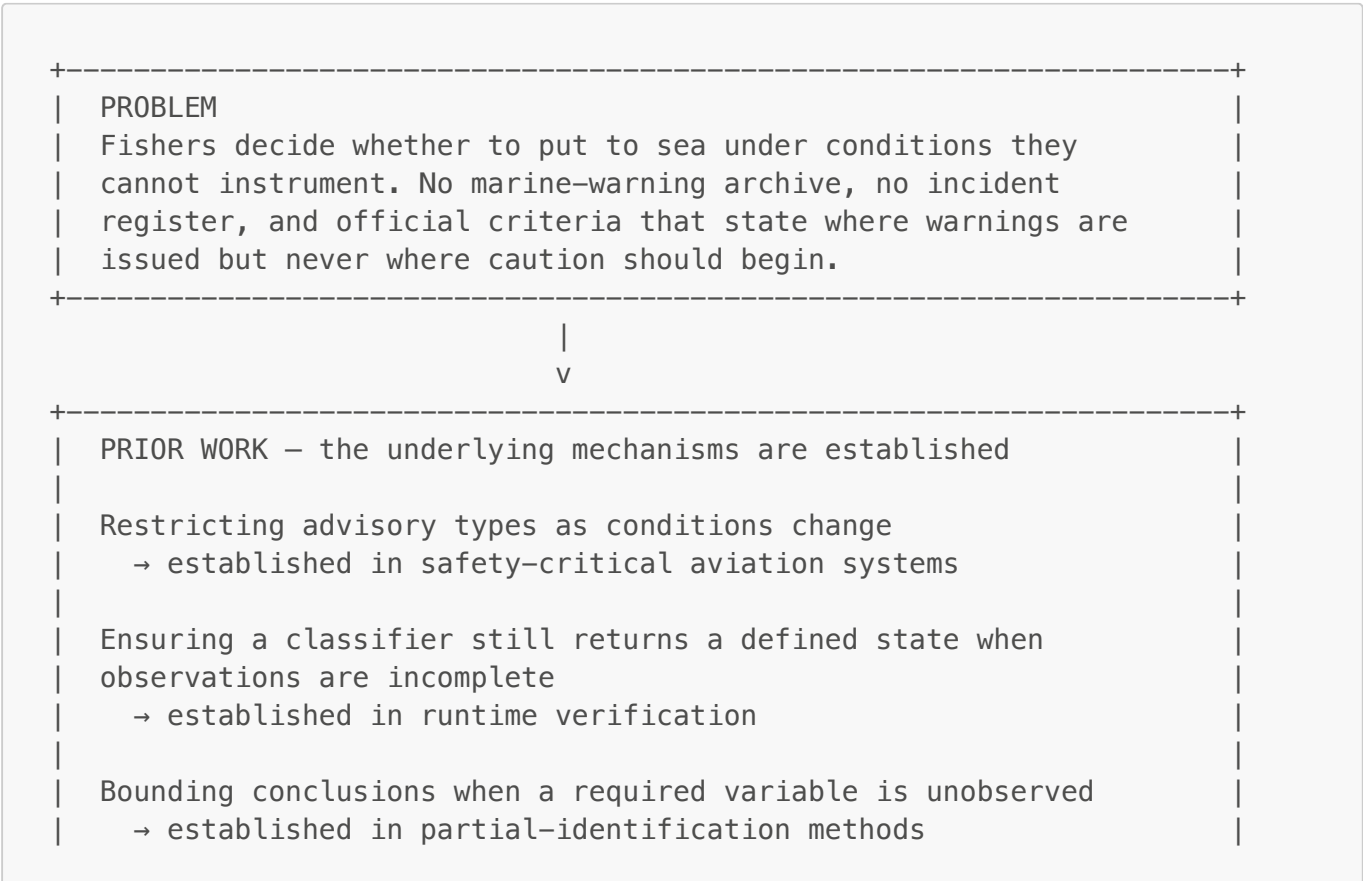
1. What this report is

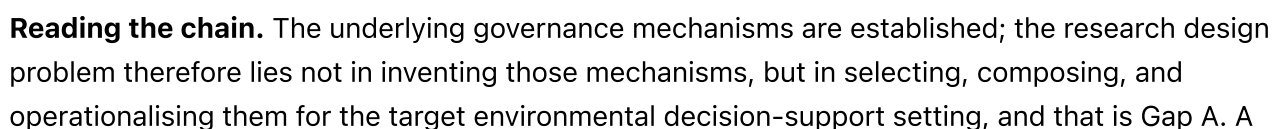
This report examines one question: where does the 5.81% come from, and what does it establish?

The experiment evaluates the isolated contribution of Level 2 advisory-scope governance by comparing the proposed graduated architecture against a participation-only gate over historical environmental conditions.

The principal experimental figures were reproduced from the canonical scripts on 2026-09-21: Δ_{L2} = 5.81% under PRIMARY and 4.48% under RESOLUTION. Historical provenance and prediction-register figures reported later in this document are drawn from their authoritative records rather than recomputed by these scripts.

2. The research chain





composition that is never exercised on real conditions would be an empty formalism, and how often it is exercised cannot be derived by proof — that is Gap B, and it is the question the replay answers. A third state label changes nothing unless it changes the admissible set, which is why C3 exists and why its zero divergence is entailed rather than measured.

3. Prior work and the gap

3.1 What is conceded

The governance mechanisms used here are established, and the thesis claims novelty for none of them.

Mechanism	Established by
Advisory-category inhibition conditioned on external state	FAA AC 20-151C (TCAS II) [notes] ; TSO-C151c (TAWS)
Human-facing advisory inhibition on observation validity	TSO-C151c §5.6 — invalid position source $\Rightarrow$ FLTA and PDA alerts inhibited, GPWS modes retained, inhibit annunciated to the crew
Multi-component classified runtime state	TCAS II: radio altitude, flap/slat/gear discretes, weight–altitude–temperature, airspeed margin, bank angle, one–engine–inoperative
Human final authority preserved	AC 20-151C §2.1.1 — <i>"TCAS II does not alter or diminish the pilot's basic authority and responsibility to ensure safe flight"</i>
Nested admissible sets, monotone contraction, weakest-link aggregation	Baxi (2026) [notes] ; TCAS II inhibit schedule
Fail-safe / fail-degraded / fail-operational regimes	Gyllenhammar et al. (2025) [notes] — ODD exit to Minimal Risk Condition
Totality of classification over incomplete observations	Bauer, Leucker & Schallhart (2011); Taleb, Hallé & Khoury (2023)
Directional bounds from declared unmeasured variables	Manski (1990, 1997); Manski & Pepper (2000)
Graduated multi-level governance; supervisory graduation	Flehmgig et al. (2024) [notes] ; Shamsujjoha et al. (2025) [notes]

3.2 Gap A — operationalisation

Advisory-type restriction conditioned on an externally measured state is established practice in certified collision-avoidance and terrain-awareness avionics; totality of classification over incomplete observations is established in runtime verification; and directional bounds from declared unmeasured variables are standard in partial-identification methodology. Within the reviewed literature, these were not found composed into a single operational specification in which participation $G(S)$ and admissible advisory scope $A_{AI}(S)$ are separately specified over a

classified multi-component environmental state, for a human-facing decision-support system in a low-resource setting where some conceptually required components are structurally unavailable.

This is a modest composition gap. The three ingredients come from unrelated fields, so their non-combination reflects disciplinary separation as much as an unexplored problem. It is not an architecture gap.

3.3 Gap B — empirical characterisation

Within the reviewed literature, no empirical characterisation was identified of how often an intermediate advisory-scope state changes the admissible recommendation set relative to a participation-only gate, when replayed over a multi-year environmental record.

This is the stronger gap and it supports the primary contribution. It could have returned a null result.

4. Research questions

RQ1 — Operational specification. Can AI participation and admissible advisory scope be specified as two separate state-conditioned functions over a classified multi-component environmental state such that classification remains total and the containment property is preserved when some conceptually required components are structurally unavailable — and which elements of that specification must be decided by the designer rather than derived from published sources?

RQ2 — Empirical characterisation and robustness. Over a multi-year historical environmental record for the deployment site, how frequently does the intermediate safety state produce a stricter admissible recommendation set than a participation-only gate, and how sensitive is that frequency to defensible environmental-threshold and data configurations?

RQ3 — Structural warrant of the multi-component classifier. Which components of the environmental state actually determine the intermediate classification at the deployment site, and is the multi-component structure warranted by the evaluated record?

Target-hardware latency is **not** a research question of this thesis. It is deferred deployment and prototype evaluation, outside the evidence required for the contribution. Contextual validation with fishers has **not** been conducted and is future socio-technical work, not a completed question.

5. Architecture

An operational architecture and specification that composes established safety-governance mechanisms — advisory-type restriction by externally measured state, totality of classification over incomplete observations, and conservative resolution of unusable inputs — for a human-facing decision-support setting in low-resource coastal fisheries. It is not a wholly novel safety architecture.

Content	
Inherited	Advisory-type restriction by external state; nested contracting admissible sets; deterministic gating outside the generator; max-severity aggregation; fail-safe

Content	
	resolution; human final authority
Domain adaptation	The avionics inhibition pattern moved from one certified platform with a validity-checked navigation input to an unattended multi-source environmental classifier; runtime-verification totality moved from trace monitoring to component-wise environmental classification; vessel-conditional ocean thresholds
Researcher-defined	$A_{AI}(CAUTION) = \{Go, Delay\}$; three-state cardinality; the policy $night \Rightarrow UNSAFE$; departure window 05:00–09:00; exclusion $D = \{m\}$; dual-configuration reporting
Empirically evaluated	Intermediate-state activation frequency; its threshold and data-configuration sensitivity; component attribution

5.1 The classifier

$f(E) = \max\text{-severity}(g_w(w), g_r(r, \kappa), g_m(m), g_o(o, v), g_t(t, date))$ — five component classifiers aggregated by worst case, with vessel category v parameterising the ocean-state thresholds rather than contributing an independent term.

Component	SAFE	CAUTION	UNSAFE	Boundary source	Status
g_w (kn, sustained)	≤ 21.6	21.6–27.0	> 27.0	MET Cat 1 / Cat 2 onsets (40 / 50 km/h)	Externally supported
g_r (mm/hr)	≤ 10.0	10.0–20.0	> 20.0	UNSAFE: MET <i>Ribut Petir</i> . CAUTION: JPS/DID <i>Light</i> limit	UNSAFE externally supported; CAUTION sourced but non-MET
g_o small (m)	< 1.0	1.0–1.25	> 1.25	Yaakob et al. (2015) [notes] operational ceiling	Externally supported
g_t	sunrise $\leq t <$ sunset	(emits no CAUTION)	otherwise	COLREGs Rule 20(b)	Boundary supported; the abstention policy is researcher-defined

Answering RQ1's second clause: four boundaries trace to named authorities; three load-bearing choices — $A_{AI}(CAUTION) = \{Go, Delay\}$, the policy $night \Rightarrow UNSAFE$, and the three-state cardinality — are decided by the designer. MET publishes no criterion below 20 mm/hr, so the rainfall CAUTION boundary is necessarily non-MET; that is a documented gap in the source material, not a concealed choice.

5.2 The governance contract

State	G(S)	A_AI(S)
SAFE	1	{Go, Delay, DepartureTime, Duration}
CAUTION	1	{Go, Delay}
UNSAFE	0	∅

A_AI(CAUTION) = {Go, Delay} is a **researcher-defined design choice**. It has a domain rationale from cited fisheries literature — Gao (2024) and Rahim et al. (2024) indicate that under marginal conditions fishers continue to make go/no-go and near-shore decisions without necessarily optimising departure timing or trip duration. **That rationale does not constitute empirical validation, optimality evidence, or evidence of improved safety.**

6. Formal properties

Four properties are proved. They establish that the implementation satisfies its intended governance contract. **None is offered as new theory.**

Property	What it means in this architecture	Status
P1 — Totality	Every valid environmental input is resolved to exactly one state: SAFE, CAUTION, or UNSAFE, including when a declared component is unavailable.	Formalisation of an established mechanism — totality over incomplete observations is established in runtime verification
P2 — Advisory-Scope Containment	As environmental severity increases, the set of advice that AI is allowed to provide can only stay the same or become smaller.	Design property — true by definition of A_AI
P3 — Safety Dominance	If the environmental state is UNSAFE, the AI advisory set must be empty; the AI cannot bypass the governance restriction.	Contract-conformance property , under assumptions A1–A4
P4 — Admissible-Set Equivalence of C1 and C3	If CAUTION allows exactly the same advice as SAFE, adding the CAUTION label alone does not change governance behaviour relative to the participation-only comparator.	Structural property — the 0.00% replay value is harness consistency, not discovery

Formal traceability. The formal results underlying P1–P3 are stated and proved in Appendix C — Formal Specification and Properties. The structural equivalence underlying P4 is established separately in the evaluation specification.

Property	Formal result	Source
P1 — Totality	Theorem C.1 (ideal domains); Theorem C.1b (operational); Corollary C.1b.1 (fail-safe)	Appendix C §C.2

Property	Formal result	Source
P2 — Advisory-Scope Containment	Theorem C.2; Corollary C.2 (strict containment)	Appendix C §C.6
P3 — Safety Dominance	Theorem C.3, under assumptions A1–A4	Appendix C §C.7; proof in §C.7.2
P4 — Admissible-Set Equivalence of C1 and C3	Structural comparator proposition (J1–P1 internally)	Evaluation Specification §6

Together, P1 and P2 establish that the classifier still returns a defined safety state and preserves the intended advisory containment even when a declared environmental component is structurally unavailable. This addresses the first part of RQ1. The mechanisms that achieve it are the declared exclusion set **D** and fail-safe resolution $g_i(\perp) = \text{UNSAFE}$: no value is fabricated for an unmeasured component, and the classifier is never left partial.

7. Evaluation

7.1 Site, data and unit of analysis

	PRIMARY	RESOLUTION
Hourly records	43,848	28,501
Period	2020-01 → 2024-12 (5.00 yr)	2021-10 → 2024-12 (3.25 yr)
Weather source	raw_weather_sea.csv (5.940246 N, 116.025 E)	same
Wave source	ERA5-Ocean, ~50 km	MFWAM, ~8 km
Departure-window denominator	9,135	5,935

Site: Kota Kinabalu, Sabah (5.98 N, 116.01 E), Open-Meteo archive. Unit of analysis: departure-window hours, 05:00–09:00, small vessel (< 10 GRT) — the deployment population.

7.2 The four conditions

Writing **FULL** = {Go, Delay, DepartureTime, Duration} and $\emptyset$ for the empty set:

	Name	A_AI(SAFE)	A_AI(CAUTION)	A_AI(UNSAFE)	Purpose
C0	Ungated	FULL	FULL	FULL	Zero point. Not a proposal
C1	Binary-gated	FULL	FULL	$\emptyset$	The comparator the contribution must differ from

	Name	A_AI(SAFE)	A_AI(CAUTION)	A_AI(UNSAFE)	Purpose
C2	Proposed	FULL	{Go, Delay}	∅	Identical to C1 except one cell
C3	Traffic-light topology	FULL	FULL	∅	Structural comparator, not a fourth arm

Fairness qualification on C3. Flehmig et al. condition their index on AI degradation — drift, outliers, performance decay — not on environmental state. C3 ports their governance *topology* onto our conditioning variable so the two structures can be compared on one axis. It is not a reproduction of their system, not a test of their architecture on their problem, and not a claim that their framework is deficient.

7.3 The identity that governs interpretation

```
A_C1  and  A_C2  agree at SAFE (FULL = FULL)
                        agree at UNSAFE (empty = empty)
                        differ at CAUTION (FULL != {Go, Delay})

therefore, for the evaluated C1/C2 mappings:

Delta_L2  =  P(S = CAUTION)
```

Verified from the implementation in `condition_comparison.py`, not from documentation, and re-confirmed numerically on every run reported below (identity holds to < 1e-9).

This is the interpretive key to the whole result. The replay measures **how frequently the intermediate restriction is activated**. It does **not** measure whether `{Go, Delay}` is the correct restriction: any strict subset of FULL at CAUTION yields the same number.

7.4 The subtraction

```
PRIMARY (43,848 hourly records, 9,135 departure-window hours)

C0 <=> C1  =  42.88%   Level 1 alone (participation gate)
C0 <=> C2  =  48.69%   Levels 1 + 2 combined
-----
difference =   5.81%   LEVEL 2 ALONE                <-- Delta_L2

Cross-check, computed independently:
C1 <=> C2  =   5.81%   (direct pairwise, same value)

Structural check (entailed by the definitions, not measured):
C1 <=> C3  =   0.00%   identical admissible sets by construction
```

8. Sensitivity

Governance mappings were held fixed throughout; only environmental classification thresholds varied.
Source: `scripts/sensitivity/threshold_sensitivity.py`.

Component	Finding
Wind	Structural zero for the evaluated site and trace. Δ_{L2} unchanged to two decimal places across <code>W_C</code> 18–25 kn. Two departure-window observations lie within ± 2 kn of 21.6; maximum 21.80 kn PRIMARY / 19.30 kn RESOLUTION
Rainfall	Minimal. <code>R_C</code> from 5.0 to 19.9 mm/hr moves Δ_{L2} by 0.48 points PRIMARY / 0.78 RESOLUTION. The non-MET lower rainfall boundary is not driving the headline result
Wave	Dominant. The wave CAUTION onset <code>O_C</code> is the principal determinant, moving Δ_{L2} by approximately –1.4 percentage points per 0.05 m over the evaluated range. Descriptive influence only — not causal

Robustness envelope, externally sourced thresholds, small vessel: 5.67% – 9.56%. Reported as a deterministic sensitivity range, **not** a confidence interval, and never written as " $5.81\% \pm x$ ".

8.1 Component attribution

PRIMARY, departure window, 531 CAUTION hours:

Component	Binding	Share
<code>g_o</code>	518 / 531	97.6%
<code>g_r</code>	13 / 531	2.4%
<code>g_w</code>	0	activates twice in 43,848 hours, decisive in neither
<code>g_t</code>	0	emits no CAUTION by construction
<code>g_m</code>	0	structurally excluded, <code>D = {m}</code>
ties	0	—

Answering RQ3, as a negative result: the five-component state remains defensible as a transferable specification, but the evaluated record does not empirically warrant equal importance of all five components at this deployment site.

Scope note. F-7's finding that three functions bind refers to max-severity binding **across all states**, where `g_t` dominates non-SAFE through darkness. The table above is **CAUTION-specific within the departure window**. Both are correct at their own scope and neither corrects the other.

9. Results

Quantity	PRIMARY	RESOLUTION
Δ_{L2} — intermediate-state activation rate (<i>empirical</i>)	5.81%	4.48%

Quantity	PRIMARY	RESOLUTION
Sourced-threshold envelope, small vessel	5.67 – 9.56%	4.26 – 6.64%
C0 ↔ C1 (Level 1 alone)	42.88%	not reported
C0 ↔ C2 (Levels 1 + 2)	48.69%	not reported
C1 ↔ C3 (structural — entailed by definitions)	0.00%	0.00%

The RESOLUTION pairwise components are withheld under `RESOLUTION_PAIRWISE_C0_C1_C0_C2 = PROVENANCE_INCOMPLETE` pending a provenance repair. `Δ_L2` itself is unaffected.

What 5.81% means. For the operative small-vessel class, the intermediate advisory-scope state was reached in 5.81% of departure-window hours under the primary configuration and 4.48% under the alternative environmental-data configuration. Because the compared conditions differ only at CAUTION (§7.3), this figure is the **activation rate of the intermediate state**. These are exact descriptive values for the analysed traces and configurations. The result is **sensitive but still informative**: it is governed almost entirely by the best-sourced threshold in the set and is non-vacuous across every sourced small-vessel configuration, but it is a point on a steep gradient rather than a stable site property.

What 0.00% is, and is not. `C1 ↔ C3 = 0.00%` follows from the comparator definitions rather than from the data, and confirms that the implementation behaves consistently with those definitions. It is **not** independent empirical evidence. The claim it supports is structural:

A three-state scheme whose intermediate state preserves the same advisory scope as the permissive state remains equivalent to the binary comparator at the admissible-set level.

Graduation comes from a change in the admissible recommendation set, not from an increase in the number of state labels.

9.1 PRIMARY versus RESOLUTION

The two configurations differ in **both** wave model and record length. A common-period comparison over the shared 28,501 hours separates them:

Effect	Value
Wave-model / data-resolution effect, record period held constant	6.45% – 4.48% = +1.97 points
Record-length effect, wave model held constant	5.81% – 6.45% = –0.64 points
Combined = reported spread	1.97 – 0.64 = 1.33 points

The two effects operate in opposite directions. The 1.33-point spread between the reported columns is therefore **not** the grid-resolution sensitivity; the like-for-like resolution effect is 1.97 points, partially offset by a record-length effect of –0.64. The direction of the earlier canonical claim — that the coarser model classifies more conservatively — is correct; its magnitude was not.

10. Contribution

Primary — empirical characterisation. An empirical characterisation, over five years of hourly environmental records for one deployment site and under two environmental-data configurations, of how often an intermediate advisory-scope state admits a strictly smaller recommendation set than a participation-only gate, together with its robustness envelope across defensible threshold configurations and an attribution of which state components produce it. This includes the negative findings: wind is a structural zero at this site, and three of five component classifiers never bind.

Secondary — domain operationalisation. A domain operationalisation for small-scale Malaysian coastal fisheries in which every threshold is traced to a named authority, the absence of an official lower criterion is documented rather than filled, and researcher-defined choices are separated from externally supported ones.

Secondary — operational specification. An operational specification in which AI participation and admissible advisory scope are expressed as two separately state-conditioned functions rather than a single graduated index.

Supporting — formal contract-conformance analysis. P1–P4 and the prototype fidelity criteria, establishing that the implementation satisfies its governance contract.

Supporting — reproducibility and provenance methodology. Predictions registered before the analysis that resolves them, refutations retained, a single authoritative figure generator, and a full supersession history.

11. Limitations and explicit non-claims

1. Δ_{L2} is invariant to the content of **A_AI (CAUTION).** Because the compared conditions differ only in that cell, the measured rate is unchanged for any strict subset of the full recommendation set. The evaluation therefore characterises how often the intermediate restriction is activated and provides **no evidence** that restricting advisory scope to **{Go, Delay}** is appropriate. That choice rests on a domain rationale drawn from the fisheries literature and is not empirically validated, shown optimal, or established as safer.

2. The result depends strongly on the wave CAUTION onset. A ± 0.25 m perturbation moves it between 2.54% and 14.75%, and the source itself supports two readings of the small-vessel UNSAFE boundary — operational ceiling and failure point — giving 5.81% and 9.22%.

3. Vessel-class specificity. The result is specific to vessels under 10 GRT. Under the same source's thresholds for medium and large vessels the intermediate state is reached in **1.96%** and **1.26%** of departure-window hours respectively (0.66% and 0.37% under RESOLUTION), approaching vacuity for the larger classes. The headline must not be generalised across vessel classes.

4. Time policy. Classifying non-daylight hours as UNSAFE is a conservative architectural policy, not a physical or legal determination; COLREGs Rule 20(b) supplies the sunrise-to-sunset boundary for navigation lights, not a requirement that AI abstain. The canonical denominator includes those hours; a daylight-only denominator gives **9.59%** and **7.40%** respectively.

5. Characterisation, not validation. No incident record exists for the site — no accident reports, no capsizings, no near-misses. The result establishes that the architecture *would have* restricted advisory scope in those hours, not that they were the correct hours to restrict.

6. **All figures are lower bounds.** The marine-warning component `m` has no historical archive and is held at `none`; the replays run with declared exclusion `D = {m}`. Since `g_m` can only raise severity, every severity figure is a lower bound. The mechanism of this inference is standard partial-identification practice.
7. **Admissible-set level only.** The metric is computed from the classifier and governance layer alone. Layer 3 exists as a research prototype with fidelity criteria closed, but the replay does not compare the *content* of generated advice.
8. **Descriptive, not inferential.** A deterministic census of every hourly record in a predefined window, not a random sample. No p-values, confidence intervals or significance tests apply.
9. **Evidential provenance.** §3 is built from the project's coding and verification documents and from primary-source verification of the closest precedents. The readings of Flehmig, Shamsujjoha and Baxi are this project's.

What this work does not establish

Accident reduction · improved safety outcomes · fisher decision quality · classification accuracy · optimal thresholds · optimal number of states · optimal `A_AI (CAUTION)` · generalisation across Sabah · generalisation across vessel classes · AI prediction accuracy · causal effectiveness · a novel fail-safe mechanism · a novel state-gating mechanism · a novel totality mechanism · a novel advisory-inhibition mechanism.

12. Provenance — why the figure moved

The headline has changed several times. Each move has a recorded cause that predates the recomputation.

Figure	Configuration	Why superseded
12.4%	Land-cell wind, ERA5 50 km waves, small-vessel UNSAFE at 1.9 m	F-10: Open-Meteo's <code>cell_selection</code> defaults to <code>land</code>
8.3%	MFWAM 8 km, UNSAFE still at 1.9 m	1.9 m was Yaakob's <i>failure point</i> , not his <i>operational ceiling</i>
6.1% (as lone headline)	MFWAM 8 km, 1.25 m	Correct for its configuration, but a 3.25-year figure reported as five years
7.84% / 6.15%	Sea-cell, <code>g_r</code> CAUTION at 7.5 mm/hr	7.5 mm/hr matched no published source; re-anchored to the JPS/DID limit
7.72% / 5.98%	Sea-cell, correct thresholds, fixed-clock <code>g_t</code>	SDR-001: the 06:00/17:00/19:00 clock had no located source
5.81% / 4.48%	Current canonical	—

Two safeguards came out of this history. **Propagating a parameter is not the same as recomputing what depends on it** — any threshold, data-source or grid-cell change now requires re-running

`canonical_figures.py` and replacing the figure table wholesale. And **predictions are registered before the analysis that resolves them**: `data/prediction-register.csv` holds P01–P24, of which P21–P24 were registered before this comparison ran, currently resolving 15 CONFIRMED / 9 REFUTED. P21 is the exception worth naming: it predicted $C1 \leftrightarrow C3 = 0$, which the comparator definitions entail, so it functions as a regression lock on the encoding rather than as a test of the site data.

13. Reproduction

```
cd <repo root>
python3 scripts/canonical_figures.py           # figure table,
both configurations
python3 scripts/condition_comparison.py        # four-condition
comparison, Delta_L2
python3 scripts/sensitivity/threshold_sensitivity.py  # sensitivity,
attribution, common period
```

The first two were re-run on 2026-09-21 (Python 3, pandas 2.3.3, numpy 2.2.6) and reproduced 5.81% / 4.48% / 0.00% / 42.88% / 48.69% exactly. The §12 provenance figures and the prediction-register counts are not produced by these scripts. Neither writes to the register by default; register writes require `--write-register`.

Inputs: `data/raw_weather_sea.csv`, `data/raw_marine_era5_sea.csv`, `data/raw_marine_mfwam.csv`, and the frozen solar artefact `data/solar/solar-events-daily.csv`. `g_t` is imported from `scripts/canonical_gt.py`; no script computes solar astronomy.

Sources

- `docs/analysis/final-research-chain-lock.md` — authoritative framing, wording and non-claims
- `docs/analysis/threshold-sensitivity-analysis.md` — sensitivity, attribution, common-period comparison
- `docs/canonical/empirical-findings-2026-09-06.md` §0a — authoritative figure table
- `publications/active/journal-1/evaluation-specification.md` — conditions, metrics, statistical treatment
- `docs/canonical/appendix-c-formalisation.md` — formal definitions and governance properties
- `data/prediction-register.csv` — prediction state authority